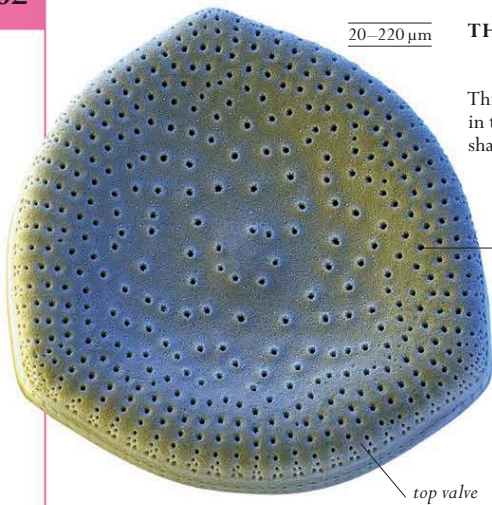


» HETEROKONTS



20–220 μm

THREE-CORNERED DIATOM*Actinopterychus* sp.

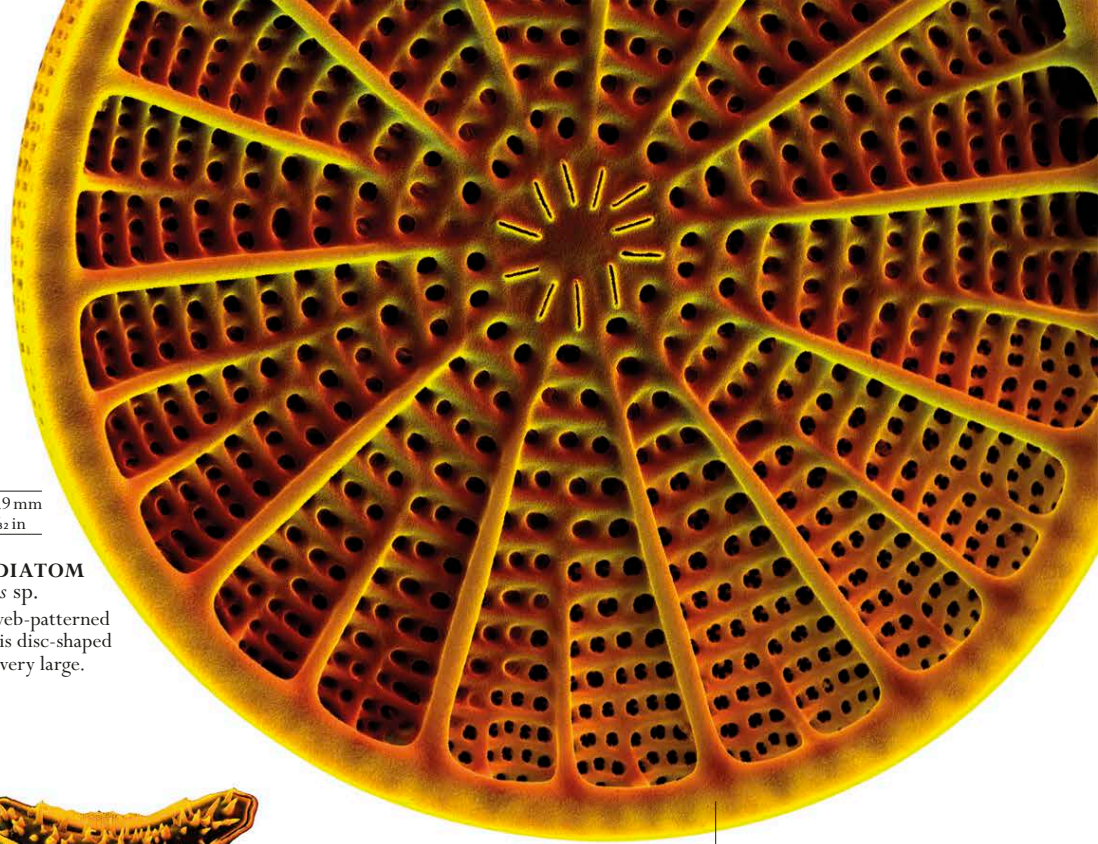
This diatom can be found in the neritic zone on the shallow continental shelf.

pores allow intake of minerals for use in photosynthesis

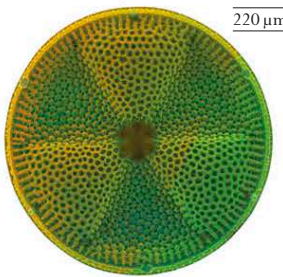
top valve

0.9 mm
1/2 in**SPIDER'S-WEB DIATOM***Arachnoidiscus* sp.

Strong radial ribs and web-patterned valves are features of this disc-shaped diatom. It can grow very large.



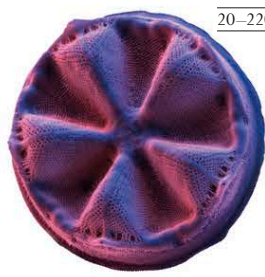
protective cell wall



220 μm

SUN-RAY DIATOM*Actinopterychus heliopenita*

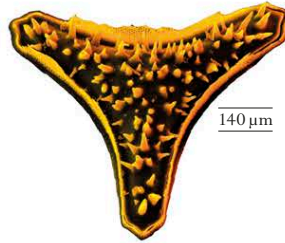
A distinctive pattern of alternating raised and depressed sectors on its top and bottom valves characterize this diatom species.



20–220 μm

FIVE-ARMED DIATOM*Actinopterychus* sp.

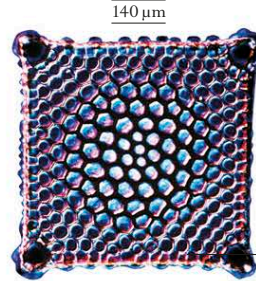
A five-arm appearance is produced by the alternating sectors on the valves of this species of the *Actinopterychus* genus.



140 μm

TRICERATIUM SP.

More than 400 species of this genus of marine diatom are known. Diatoms of this genus are often, but not always, three sided.

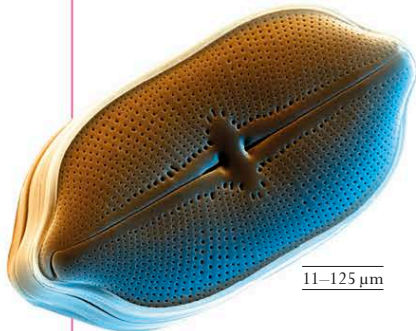


140 μm

TRICERATIUM FAVUS

The heavily silicified cell walls of this species can leave a trail of silica in the water as it moves. This can show how seawater penetrates into freshwater environments.

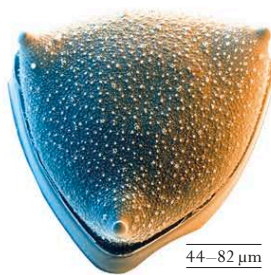
geometric cell wall



11–125 μm

BOAT-SHAPED DIATOM*Navicula* sp.

This species belongs to the largest genus of diatoms, which has thousands of species.



44–82 μm

DAPPLED DIATOM*Stictodiscus* sp.

Seen under a scanning electron microscope, this diatom shows pores dappled on its valve surface, above a surrounding girdle.



shrub-like fronds

THONGWEED*Himantalia elongata*

During its reproductive phase, this brown seaweed of the northern hemisphere produces long strap-like fronds, which are types of divided leaves.

60 cm
23 1/2 in**TOOTHED WRACK***Fucus serratus*

This robust, shrunken, olive-brown seaweed grows on lower shores all round the North Atlantic.

4 m
13 ft**SUGAR KELP***Saccharina latifolia*

Found on rocky shores and at depths of 8–30 m (26–100 ft) in northern seas, this brown seaweed is also often known as sugar wrack.

2–3.5 m
6 1/2–11 ft**CUVIE***Laminaria hyperborea*

This alga is commercially important in iodine production. It grows at depths of 8–30 m (26–100 ft), mainly in the northern hemisphere.

fronds are leaf-like structures used in photosynthesis

air bladder on frond helps keep seaweed buoyant

1–3 m
3 1/4–10 ft**WIREWEED***Sargassum muticum*

Originally from Japan, this invasive brown seaweed has now spread to Europe. It can grow as much as 10 cm (4 in) in a day.

30–100 cm
12–39 in**SEA OAK***Halidrys siliquosa*

A large brown seaweed, common in rock pools in Europe, this species has a distinctive zigzag stem and air-filled pods known as air bladders.